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Author(s): N. Kakwani and K. Subbarao

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Rural Poverty and Its Alleviation in India

N Kakwani

K Subbarao

To understand the impact of economic growth on poverty, it is important to measure separately the impact of changes in average income and income inequality on poverty. This paper makes an attempt in this direction. It examines past trends in the distribution and growth of consumption and assesses their relative impact on the poor and the ultra poor, over time and across states. The paper addresses the following questions: How has the distribution of per capita consumption changed over time across the states, and to what extent it has nullified the beneficial impact of growth on poverty? How have the poor and the ultra poor fared in the wake of changes in consumption and its distribution? What have been the regional patterns and to what extent can these patterns be explained by our knowledge of the characteristics and public interventions in operation in different regions/states? The authors consider the potential indirect role of agriculture and manufacturing, as well as the contribution of direct anti-poverty programmes that have been in operation for over a decade now and the trends in social sector expenditures and their outcomes.

MUCH has been written about poverty in India over the last two decades. In the initial years considerable attention was devoted to the estimation of poverty. More recently, serious attempts have been made to analyse the facts explaining poverty across regions and over time. While controversy still prevails over both estimation and explanation, several insights have been gained, the foremost among them being the significant impact of a faster rate of growth of agricultural output per head on the incidence of poverty.

In this debate on poverty in India, the two issues that attracted much attention are: (a) whether or not economic growth contributed to a reduction in the incidence of poverty (i.e., the existence or otherwise of the so-called 'trickle-down' mechanism), and (b) whether or not economic growth has been accompanied by an increase in inequality of income or consumption. In the past, analysts have examined these issues by looking at the distribution of assets (land) or income or consumption in the high and slow-growing districts/states, often using single-year cross-section data. To understand the impact of economic growth on poverty, it is important to measure separately the impact of changes in average income (consumption) and income (consumption) inequality on poverty, over time. This paper makes an attempt in this direction in that it examines past trends in the distribution of growth of consumption and assesses its relative impact on the poor and the ultra poor, over time and across states of India.

Using the National Sample Survey data for fifteen major states of India for 1972-73, 1973-74, 1977-78 and 1983, the paper addresses the following questions.

How has the distribution of per capita consumption changed over time across the states, and to what extent has it nullified the beneficial impact of growth on poverty? How have the poor and ultra poor fared in the wake of changes in consumption and its distribution? What have been the regional patterns and to what degree have these patterns been explained considering our

knowledge of the structural characteristics of these states and the state-wise performance indicators of anti-poverty programmes?

This is followed by an analysis of India's poverty alleviation strategies: The potential indirect role of agriculture and manufacturing, as well as the contribution of direct poverty alleviation interventions are analysed, against the backdrop of the results pertaining to poverty and inequality.

In particular, we examine the extent to which the regional allocation of funds for direct poverty alleviation programmes is sensitive to the regional distribution of the poor and the ultra poor, and whether the recent evidence on the impact of these programmes is consistent with the observed patterns in poverty and inequality.

So far the debate on poverty has focused exclusively in terms of 'income' or 'expenditure' needed to command the critical minimum basket of goods. This concentration of attention on 'income' has resulted in the neglect of other influences—notably education and health—which expand and ensure the long-run capabilities of the population.

Concluding sections of this paper deal with the trends in social sector expenditures and their outcomes, and assesses the performance of different states in relation to their perceived needs. The last section ties together the main findings and draws some inferences for policy.

I

Methodology

To analyse poverty, we need to measure the economic welfare of each individual in the society. Although income is widely used to measure economic welfare, it has many serious drawbacks.¹ One major drawback of using income as a measure of economic welfare is that it may have substantial fluctuations which are averaged out in the long run. Therefore, it has been suggested that consumption is a better indicator of the actual economic position of a household

than its current income.

The National Sample Surveys provide a reasonably comparable time series data on the levels and distribution of household consumption expenditures. These data are available in grouped form, giving for each group: (a) the percentage distribution of an estimated number of persons, and (b) the average consumer expenditures in rupees per person. The monthly per capita expenditure levels are generally grouped into 12 to 14 expenditure classes. To estimate poverty from such data, one needs to employ some interpolation device. A commonly used procedure is to fit a density function to the entire consumption range and then compute poverty measures from the parameter of the fitted function. Most of the Indian studies have employed a two-parameter lognormal distribution [Minhas, Jain, Kansal and Saluja, 1987]. The difficulty with this approach is that the lognormal distribution tends to overcorrect the positive skewness of the income distribution and thus fits poorly to the actual data.

In the present paper we used a general interpolation device proposed by Kakwani [1980]. This method utilises, within each expenditure range, a separate, continuous differentiable function which exactly fits to the data points. The inequality and poverty measures are then computed by linking this function. We used a polynomial of the third degree to represent the Lorenz curve within each income class, except the first and the last open-ended classes. For the first and last expenditure ranges, a Pareto curve is used as a further refinement.

In this paper we have used per capita household expenditure as a measure of household economic welfare. Expenditure comprises all expenditures incurred by the household exclusively on domestic account including consumption out of home-grown produce or transfer receipts like gifts, loans, etc. The expenditure on household enterprises is excluded from consumer expenditure. Consumption out of home-grown produce is evaluated at ex-farm rates. Ideally, one should include the depreciated value of

consumer durables which are in stock in the household. Because of non-availability of information on stock values, the monetary value of consumer durables acquired during the reference period is included in the total expenditure. This may in some cases distort the results on income inequality.

When the index of household welfare is constructed, the next step involves the determination of the welfare of the individuals in the households. In this paper individual welfare was derived by assigning every individual in a household a welfare value equal to the per capita consumption for that household. The validity of this assumption is discussed in Kakwani [1986].

Once we have decided upon a suitable index of economic welfare for individuals, the next step is to find a threshold welfare level below which an individual is poor. In the present paper we have used two poverty lines. The Planning Commission [1979] has defined poverty threshold as the per capita monthly expenditure of Rs 49.09 in rural India at 1973-74 rural prices. Following the Planning Commission, we adopted a round figure of Rs 50 as our first poverty line. This poverty line corresponds roughly to the per capita daily requirement of 2,400 calories in rural areas. Our second poverty line is equal to the per capita consumption of households of less than 80 per cent of the Planning Commission's poverty threshold of Rs 50. We define the households whose per capita consumption is below Rs 40 per month as 'ultra-poor'; those whose physical personal maintenance is unstable [Lipton 1988]. The ultra-poor cut off point of Rs 40 corresponds closely to the poverty line used by Bardhan [1970] and Dandekar and Rath [1971], which is Rs 15 at 1960-61 prices (this line at 1973-74 rural prices is equivalent to the per capita expenditure of Rs 42.5).

To compare poverty across different time periods, one needs to adjust the distributions given in current prices for price changes over time. Minhas [1970] and Dandekar and Rath [1971] used the national income deflator, which has been criticised by Bardhan [1974]. This deflator does not reflect the prices which people actually pay for their consumption goods. We have followed Bardhan in using the Consumer Price Indices for Agricultural Labourers (CPIAL) prepared by the Labour Bureau. This index, constructed on the basis of the monthly retail prices of 75 consumer items collected from selected rural centres, uses the consumption pattern of rural agricultural households observed in 1956-57. A major objection that has been raised against this index is that the agricultural labour households constitute only about 30 per cent of the total rural population and the remaining 70 per cent of the rural households which constitute a large number of poor small farmers may have a quite different consumption pattern [Minhas et al 1987]. This objection is, of course, relevant but Bardhan [1974] has observed that the consumption patterns of agricultural labour households are not very

different from those of the other household groups in the rural areas.

Recently, Minhas et al [1987] have constructed a new price index which depicts the movement of consumer prices for the entire rural population. An attractive feature of this index is that it is based on the consumption patterns observed in more recent years than the CPIAL. This index is not available as yet for individual states.

Minhas et al [1987] have applied the index to compute the incidence of poverty in the entire rural and urban areas of India for 5 different periods ranging from 1970-71 to 1983. They performed the poverty computations on the All India expenditure distributions which are published by the NSS separately for rural and urban areas. These aggregate distributions are derived from the state distributions by means of population weighted averages. This procedure of obtaining the aggregate distributions is not satisfactory because of widely observed differences in price levels in different states. The

expenditures in current prices given for each state have to be adjusted for state-wise price differences before they can be aggregated. Since this procedure was not adopted in the Minhas et al study, their All India estimates of rural poverty are inaccurate notwithstanding the advance made on the construction of a new price index.

To obtain comparable poverty levels across different states, we need relative price levels in the rural areas of different states in a given year. Bhattacharya and Chatterjee [1974] worked out these state-wise price relatives for the year 1963-64, which also formed the basis for the Bardhan [1973] study. Using these price relatives and the CPIAL available at the state level we worked out the state-specific poverty lines at the current prices for the years 1972-73, 1973-74, 1977-78 and 1983.² The results are presented in Table 1.

It can be seen from Table 1 that the poverty line varies substantially across the states as well as over time. These differences are attributable entirely to the variations in

TABLE 1: STATE-SPECIFIC POVERTY LINES FOR DIFFERENT NSS ROUNDS

	Poor				Ultra Poor			
	1972-73	1973-74	1977-78	1983	1972-73	1973-74	1977-78	1983
Andhra Pradesh	37	44	54	76	30	35	43	14.5
Assam	42	50	61	99	33	40	48	79
Bihar	44	62	63	104	35	50	50	83
Gujarat	44	48	56	85	35	38	44	68
Haryana	42	51	62	92	34	40	49	73
Karnataka	40	50	56	90	32	40	45	72
Kerala	41	52	59	100	33	41	47	80
Madhya Pradesh	40	51	57	85	32	41	46	68
Maharashtra	46	52	60	91	36	41	48	72
Orissa	37	48	52	94	29	38	42	75
Punjab	42	51	62	92	34	40	49	73
Rajasthan	38	49	56	78	30	39	45	62
Tamil Nadu	37	47	59	99	30	37	47	79
Uttar Pradesh	40	51	57	87	32	41	45	70
West Bengal	46	59	68	112	37	47	54	89
All India	40	50	57	91	32	40	46	72

TABLE 2: INEQUALITY AND POVERTY: ALL INDIA RURAL *

Indicators	1972-73	1973-74	1977-78	1983
Per capita real consumer expenditure at 1973-74 prices	53.0	51.8	56.7	61.7
<i>Theil's inequality</i>				
Within state inequality	14.2 (92)	12.3 (93)	15.1 (91)	13.8 (88)
Between state inequality	1.3 (8)	0.9 (7)	1.5 (9)	1.8 (12)
Total inequality	15.5 (100)	13.2 (100)	16.6 (100)	15.6 (100)
<i>Poor</i>				
Head-count ratio (per cent)	60.5	60.5	56.3	48.4
Number of poor (millions)	264.2	269.1	269.3	255.4
Poverty gap ratio	19.5	18.8	17.5	13.8
Watts measure	27.0	26.7	23.2	18.9
<i>Ultra-poor</i>				
Head-count ratio (per cent)	42.1	41.3	37.9	30.3
Number of poor (millions)	183.8	183.4	181.1	159.8
Poverty gap ratio	11.3	10.9	9.9	7.3
Watts measure	14.9	14.4	12.5	9.4

* Figures in the bracket give the per cent contribution of within and between states inequality.

the prices. For instance, West Bengal has the highest poverty line for each of the 4 years; it means that West Bengal is the most expensive state to live. It also had the highest inflation rate particularly over the 1977-78 to 1983 period.

The incidence of rural poverty for each state was first computed using the state-specific poverty lines given in Table 1. The poverty for all India rural level was then derived from the rural poverty levels computed at the state levels. The trends in inequality and poverty at the all India rural level are discussed in the next section.

II

Trends in Inequality and Poverty: All India Rural

Having decided upon the poverty line, the next step is to compute a poverty index which would adequately measure the intensity of poverty. The head-count measure, which is defined as the proportion of individuals who have income below the poverty line, is widely used to analyse poverty. This is a crude poverty index because it does not take account of the income-gap among the poor. If the degree of misery suffered by an individual is proportional to the income shortfall of that individual from the poverty line, then the sum total of these shortfalls may be considered an adequate measure of poverty. Such a measure is called the poverty gap ratio and can be written as:

$$g = \frac{\int_0^z g(x)f(x)dx}{z} = \frac{H(z - \mu^*)}{z} \quad \dots (1)$$

where $g(x) = \frac{(z-x)}{z}$, z being the poverty line,

$f(x)$ is the density function of income x , H is the head-count ratio and μ^* is the mean income of the poor.

The measure g will provide adequate information about the intensity of poverty if all the poor are assumed to have exactly the same income, which is less than the poverty line. In practice, the income among the poor is unequally distributed and, therefore, g cannot be an adequate measure of poverty. To make g sensitive to the income inequality among the poor Sen [1976] proposed a poverty measure which led to a large theoretical literature on the measurement of poverty.³ The main difficulty with Sen's measure is that it is not additively decomposable. It is, therefore, awkward to compute its value for the all India rural population using the state level rural expenditure distributions. Moreover, the additively decomposable poverty measures are useful because they allow assessment of the effects of changes in sub-group poverty to total poverty.

In 1968, Watts proposed an additively decomposable poverty measure which can be obtained by substituting $g(x) = (\log z - \log x)$ in equation 1:

$$W = \frac{1}{\mu^*} \int_0^z (\log z - \log x) f(x) dx \quad (2)$$

Although this is not a well known measure, at the same time, it is simple to compute and has all the important attributes: it satisfies Sen's monotonicity and transfer axioms and also Kakwani's [1986] transfer-sensitivity axiom.

Watts' measure is also closely related to income inequality. If we substitute $z = \mu$, μ being the mean income of the entire population and evaluate the integral (equation 2) over the whole range of x , we obtain:

$$T = \int_0^\infty (\log \mu - \log x) f(x) dx \quad (3)$$

which is one of Theil's [1967] two well known inequality measures. An attractive feature of this measure is that it can be expressed as the sum of the between and within group inequalities.

In the present exercise we have used only the additively decomposable poverty and inequality measures. These measures have been employed to derive poverty and inequality in each of the 15 major states of India. The results obtained from individual states have then been aggregated to obtain the all India situation.⁴

Table 2 presents the empirical results on inequality and poverty measures aggregated over the rural areas of 15 major states. The conclusions emerging from this table are summarised below.

It is interesting that the average consumption per capita fell by 2.3 per cent in 1973-74 (a normal crop year) from the previous year, but rose impressively by 2.3 per cent per annum over the period 1973-74 to 1977-78 and modestly at 1.5 per cent over the period 1977-78 to 1983.

The total inequality in per capita consumption (measured by Theil's index) fell in 1973-74 modestly, rose dramatically by 1977-78 but fell again in 1983. Thus, in general, during the periods of high growth rates, the inequality has shown a tendency to increase.

The between state inequality contributed only eight per cent to the total inequality in

1972-73, which means that the within-state inequality is the major determinant of the inequality at the all India level. Therefore, policies aimed at redistribution of income and assets within the poorer states such as Bihar undoubtedly deserve greater emphasis. However, the share of between state inequality has grown monotonically from eight per cent to twelve per cent between 1972-73 and 1983. Although the within states inequality fell rapidly between 1977-78 and 1983, the between state inequality remained more or less at the same level. Thus, the regional disparities in average living standards have shown a tendency to increase over time.

As for poverty ratios, the ratio for ultra poor fell to a greater extent than for the poor. Between 1972-73 and 1973-74, there was no change in the head-count ratio for the poor but that for the ultra poor fell by about one per cent. It appears that a favourable crop benefits the ultra poor even if the overall poverty ratio may not fall. Over the period 1973-74 to 1977-78, poverty ratio

TABLE 4: RANKING OF STATES BY PER CAPITA REAL CONSUMPTION

	Poor			
	1972-73	1973-74	1977-78	1983
Andhra Pradesh	10	12	12	12
Assam	5	8	4	5
Bihar	3	3	2	1
Gujarat	12	11	11	10
Haryana	14	14	13	14
Karnataka	11	9	8	9
Kerala	8	10	10	11
Madhya Pradesh	7	4	6	6
Maharashtra	2	6	3	8
Orissa	4	2	5	3
Punjab	15	15	15	15
Rajasthan	13	13	14	13
Tamil Nadu	6	7	7	4
Uttar Pradesh	9	5	9	7
West Bengal	1	1	1	2

TABLE 3: TRENDS IN AVERAGE CONSUMPTION PER CAPITA (REAL) BY STATES, 1972-1983

States	Per Capita Consumption (at 1973-74 Prices)				Annual Growth Rates		
	1972-73	1973-74	1977-78	1983	1973-74 Over 1972-73	1977-78 Over 1973-74	1983 Over 1977-78
					Per Cent	Per Cent	Per Cent
Andhra Pradesh	53.8	58.3	65.1	75.9	8.4	2.8	2.8
Assam	50.2	52.0	49.2	57.4	3.6	-1.4	2.8
Bihar	46.8	45.2	45.9	45.3	-3.4	0.4	-0.2
Gujarat	58.8	56.8	63.3	72.6	-3.4	2.7	2.5
Haryana	83.5	71.8	75.1	83.0	-14.0	1.1	1.8
Karnataka	56.6	52.3	57.9	65.3	-7.4	2.6	2.2
Kerala	51.5	53.8	63.0	73.0	4.5	4.0	2.7
Madhya Pradesh	51.5	49.3	52.5	59.5	-4.3	1.6	2.3
Maharashtra	45.7	50.8	48.7	61.0	11.2	-1.0	4.2
Orissa	47.9	44.9	50.5	52.8	-6.3	3.0	0.8
Punjab	88.8	74.8	93.0	93.2	-15.8	5.6	0
Rajasthan	69.3	66.0	75.3	81.3	-4.8	3.4	1.4
Tamil Nadu	50.9	51.3	53.6	56.7	0.8	1.1	1.0
Uttar Pradesh	53.3	50.3	59.6	60.1	-5.6	4.3	0.2
West Bengal	42.0	40.6	43.6	46.9	-3.3	1.8	1.3
All India	53.0	51.8	56.7	61.7	-2.3	2.3	1.5

(head-count) for the poor fell at an annual rate of 1.8 per cent. A similar pattern is noticeable in the next period 1977-78 to 1983 when head-count ratio for the poor fell by 2.9 per cent, but by 4.4 per cent for the ultra poor.

The trends in poverty gap ratio and Watts measures are similar. In general, the reduction in poverty is more for the ultra poor than for the poor, and the magnitudes of reduction in poverty are higher for the period 1977-78 to 1983 than for the former period 1973-74 to 1977-78. This is an interesting result because 1973-74 to 1977-78 was a period of higher growth (2.7 per cent) with increasing inequality whereas the 1977-83 period was characterised by a somewhat lower per capita growth of consumption (1.7 per cent) but accompanied by a substantial decrease in inequality. The decrease in inequality was the major factor which led to a substantial reduction in poverty in the second period.

While no significant changes in the *pattern* of growth would be observed between the two periods, the second period (1977-78 to 1983) witnessed a spate of direct anti-

poverty interventions which were basically aimed at improving the income and consumption levels of the low-income households. If this strategy were working with reasonable efficiency, one would expect a reduction in consumption inequality in the second period compared to that in the first period (1973-74 to 1977-78). A significant reduction in inequality in the second period compared to that in the first period is quite noticeable from the empirical results presented in Table 2. An important question arises: whether the improvement in inequality can be attributed to the anti-poverty programmes which started around 1977-78. An attempt will be made to answer this question in the next section where we will analyse poverty trends at the state level.

III State-wise Trends in Average Consumption Per Capita and Inequality

State-wise trends in average consumption per capita are shown in Table 3. Over the one year initial period (1973-74 over 1972-73),

TABLE 5: STATE-WISE PATTERNS IN INEQUALITY IN CONSUMPTION, 1972-73 TO 1983

	Gini Index				Theil's Measure			
	1972-73	1973-74	1977-78	1983	1972-73	1973-74	1977-78	1983
Andhra Pradesh	0.28	0.29	0.30	0.295	0.127	0.143	0.151	0.142
Assam	0.19	0.21	0.18	0.198	0.062	0.075	0.057	0.064
Bihar	0.29	0.28	0.26	0.260	0.141	0.128	0.112	0.109
Gujarat	0.31	0.24	0.29	0.260	0.127	0.094	0.136	0.105
Haryana	0.29	0.30	0.30	0.280	0.135	0.142	0.147	0.123
Karnataka	0.28	0.29	0.32	0.300	0.129	0.131	0.174	0.150
Kerala	0.31	0.32	0.36	0.330	0.163	0.167	0.212	0.189
Madhya Pradesh	0.31	0.29	0.34	0.300	0.159	0.137	0.188	0.143
Maharashtra	0.32	0.27	0.30	0.290	0.172	0.123	0.143	0.133
Orissa	0.32	0.27	0.31	0.270	0.162	0.117	0.153	0.119
Punjab	0.32	0.28	0.31	0.290	0.163	0.127	0.165	0.135
Rajasthan	0.32	0.28	0.33	0.340	0.170	0.131	0.184	0.196
Tamil Nadu	0.28	0.28	0.32	0.330	0.129	0.125	0.174	0.177
Uttar Pradesh	0.28	0.24	0.30	0.290	0.131	0.095	0.151	0.141
West Bengal	0.31	0.30	0.30	0.290	0.156	0.148	0.143	0.138

TABLE 6: ANNUAL PERCENTAGE CHANGE IN INEQUALITY

	Gini Index			Theil's Measure		
	1972-73	1973-74	1977-78	1972-73	1973-74	1977-78
	to	to	to	to	to	to
	1973-74	1977-78	1983	1973-74	1977-78	1983
Andhra Pradesh	5.4	0.7	-0.4	12.6	1.3	-1.1
Assam	10.0	-3.1	1.3	21.0	-6.6	2.1
Bihar	-5.1	-1.6	-0.2	-9.2	-3.3	-0.5
Gujarat	-22.3	4.8	-2.2	-26.0	9.7	-4.6
Haryana	3.8	-0.3	-1.1	5.2	0.9	-3.2
Karnataka	2.9	3.2	-1.3	1.6	7.4	-2.7
Kerala	1.9	2.8	-1.3	2.5	6.1	-2.1
Madhya Pradesh	-6.1	3.5	-2.3	-13.8	8.2	-4.9
Maharashtra	-14.8	2.2	-0.5	-28.5	3.8	-1.3
Orissa	-15.5	3.4	-2.3	-27.8	6.9	-4.5
Punjab	-11.7	3.1	-1.6	-22.1	6.8	-3.6
Rajasthan	-12.1	3.8	0.7	-22.9	8.9	1.2
Tamil Nadu	1.1	4.1	0.2	-3.1	8.0	0.3
Uttar Pradesh	14.4	5.8	-0.6	-27.5	-0.9	-0.6
West Bengal	-2.9	-0.4	-0.5	-5.1	-0.9	-0.6
All India	—	—	—	-14.8	5.9	-1.1

average consumption fell in most states. The notable exceptions are Andhra Pradesh, Assam, Kerala, Maharashtra and Tamil Nadu. Over the period 1973-74 to 1977-78, all states except Assam have shown a positive trend. The positive trend continued in the following period as well as in all states except Bihar, Maharashtra, Rajasthan and Punjab.

It is interesting to note that the two richest states Haryana and Punjab suffered the largest decline in the per capita real consumption during the 1972-73 to 1973-74 period. But in the subsequent periods, both states maintained a positive growth in per capita consumption. The two poorest states, West Bengal and Bihar also suffered a substantial decline in per capita consumption in the first period but in the second and subsequent period, they managed to arrest further decline.

To see if the level of per capita consumption is related to the growth rate, we estimated the following regressions:

- 1972-73 to 1973-74 :
(GPCC) = $18.7 - 0.38$ (PCC), $R^2 = 0.47$ (0.11)
- 1973-74 to 1977-78 :
(GPCC) = $1.9 + 0.07$ (PCC), $R^2 = 0.14$ (0.05)
- 1973-74 to 1983 :
(GPCC) = $2.8 - 0.02$ (PCC), $R^2 = 0.04$ (0.03)

where

PCC : Per capita consumption
GPCC: Growth rate in per capita consumption

The above relations suggest that in the 1972-73 to 1973-74 period, the growth rate tended to be inversely correlated with the level of per capita consumption, or in other words, the richer states tended to have a lower growth rate. In the subsequent periods the relationship between the growth rate and the level of per capita consumption was not found to be statistically significant.

Table 4 presents the ranking of states by per capita consumption. The results show that the states which have substantially improved their relative positions between 1972-73 and 1983 are Maharashtra and Kerala notwithstanding slower growth of agricultural output in both states. This is not surprising, in view of the past record of these states in poverty alleviation interventions (Maharashtra's Employment Guarantee Scheme, and Kerala's long-standing public distribution system). The states whose relative position has deteriorated during the same period are Uttar Pradesh, Bihar, Gujarat, Karnataka, and Tamil Nadu. The remaining states maintained more or less the same ranking during the periods considered.

Let us now turn to trends in inequality in per capita consumption. These are shown in Tables 5 and 6. Table 5 presents the levels of inequality and Table 6, the annual percentage change.

The results show a wide variation in inequality across states as well as over time. In 10 out of 15 states, the inequality declined

during the 1972-73 to 1973-74 period. The decline was highest in Gujarat (22.3 per cent), followed by Orissa (15.5 per cent) and Maharashtra (14.8 per cent). Assam registered a substantial increase in inequality. This situation changed dramatically during the 1973-74 to 1977-78 period, during which 11 out of 15 states showed an increase in inequality.

After looking at NSS published data we noted that in Maharashtra and Rajasthan in 1977-78 there was a very large increase in expenditures on durables for the highest expenditure class. The expenditure levels were unacceptably high. Consultations with the NSS office confirmed that these were errors. We, therefore, adjusted the numbers to conform broadly to the temporal changes in the same states in the years immediately preceding and following 1977-78. The inequality estimates shown in Tables 5 and 6 reflect these adjustments. It is worth stressing, however, that since these unacceptably high values occurred only in the top expenditure range, they would have little effect on the poverty estimates.

It is quite evident that inequality increased substantially in most states between 1973-74 and 1977-78, but this situation again changed in the subsequent period. Inequality declined in 12 out of 15 states during this period. Assam, Rajasthan and Tamil Nadu were the only states which showed a slight increase in inequality.

As noted earlier, the widespread decline in inequality in the second period could be due to the major *policy change* that took place in the late 1970s related to the introduction of various anti-poverty programmes whose main objective was to raise the consumption levels of the poor. It is possible that this policy contributed towards an overall reduction in consumption inequality especially in the states with better rural administration.

The past research on poverty has shown that the movement in poverty is related to the movement in foodgrain production. Ahluwalia [1978] demonstrated a statistically significant inverse relationship between rural poverty and agricultural performance for India as a whole, suggesting that agricultural growth by itself tends to reduce the incidence of poverty. His analysis of the individual states, however, did not provide a conclusive evidence of this proposition. This issue clearly needs further investigation but a natural question that arises from it is whether the reduction in poverty is brought about by an increase in per capita consumption or reduction in the inequality of consumption, or some combination of the two. From the agricultural production point of view, 1983 was an excellent year. Foodgrain production increased from 129.4 million tonnes in 1982 to 152.4 million tonnes in 1983. The hypothesis which is yet to be tested is whether the bumper food production leads to a reduction in the inequality of per capita consumption. If this hypothesis is valid, then the widespread reduction in inequality between 1977-78 and 1983 could also be attributed to the highly favourable foodgrain production in 1983.

Finally in this section we would like to investigate whether there is a significant relationship between growth rates in per capita consumption and percentage changes in the inequality of consumption. To do this, we estimated the following regressions.

1 1973-74 to 1977-78:

$$GGI = 0.38 + 0.82 (GPC), R^2 = 0.42$$

$$(0.29)$$

2 1977-78 to 1983:

$$GGI = 0.86 + 0.03 (GPC), R^2 = 0.001$$

$$(0.24)$$

where

GPC = Growth rate of per capita consumption

GGI = Growth rate of the Gini index

The relations suggest that the growth rate in consumption per capita is positively and significantly related to the growth rate in the Gini index. It means that the faster growth rate in consumption tends to introduce greater inequality.⁵ The equation relating to the first time period (1973-74 to 1977-78) indicates the stronger impact of growth on in-

equality than that in the second period. It is interesting that the adverse impact of growth on inequality is to be observed only in the first period, but not in the second period. As argued above, although the factors that may have contributed to the virtual absence of this relationship in the second period are difficult to establish empirically, it is at the same time difficult to dismiss entirely the role of anti-poverty interventions introduced in this period.

IV

Regional Trends in Poverty

The regional trends in poverty for the poor and ultra poor are shown separately in Tables 7 and 8, respectively. Each of these tables presents results on both head-count and poverty-gap ratios. The numerical results of Watts' measure are presented in Table 1 of the Appendix.

For the 'poor', the annual change over the first two years 1972-73 and 1973-74 was negative for Andhra Pradesh, Assam, Gujarat, Kerala and Maharashtra; almost no

TABLE 7: TRENDS IN HEAD-COUNT RATIO AND POVERTY GAP RATIO FOR POOR BY STATES, 1972-73 TO 1983

	Head-Count Ratio				Poverty Gap			
	1972-73	1973-74	1977-78	1983	1972-73	1973-74	1977-78	1983
Andhra Pradesh	56.4	53.7	43.6	30.2	16.4	14.8	11.6	6.7
Assam	59.6	54.9	62.1	43.7	13.4	13.0	13.9	8.3
Bihar	67.3	70.6	69.0	70.2	23.5	24.1	22.6	23.0
Gujarat	53.8	48.5	44.0	28.5	15.2	11.7	11.9	5.3
Haryana	23.2	37.7	31.1	20.8	4.6	9.1	7.2	4.0
Karnataka	52.5	60.0	54.0	42.6	15.3	18.5	17.3	12.1
Kerala	63.6	60.9	51.7	39.6	21.1	20.4	16.8	9.9
Madhya Pradesh	64.7	64.9	64.0	51.3	21.3	21.4	21.4	14.2
Maharashtra	75.4	61.9	66.8	48.2	26.6	19.0	22.4	12.7
Orissa	68.5	69.5	63.7	56.2	24.8	24.4	21.1	16.7
Punjab	22.6	30.0	19.0	15.7	4.8	6.6	4.2	2.9
Rajasthan	40.9	40.9	34.1	31.7	11.4	10.3	9.4	8.3
Tamil Nadu	61.6	61.8	61.1	56.3	18.7	18.8	19.8	18.1
Uttar Pradesh	58.3	60.4	52.5	49.4	17.8	17.0	14.6	13.6
West Bengal	75.6	77.6	73.4	67.9	29.8	31.1	27.1	23.3
All India	60.5	60.5	56.3	48.4	19.5	18.8	17.5	13.8

TABLE 8: TRENDS IN HEAD-COUNT RATIO AND POVERTY GAP RATIO FOR POOR BY STATES, 1972-73 TO 1983

States	Head-Count Ratio				Poverty Gap			
	1972-73	1973-74	1977-78	1983	1972-73	1973-74	1977-78	1983
Andhra Pradesh	37.5	31.7	25.1	14.5	9.2	10.9	5.8	2.9
Assam	28.6	26.8	30.1	17.9	4.9	5.8	5.0	2.7
Bihar	49.5	52.4	48.7	50.3	14.4	15.0	13.1	13.4
Gujarat	34.2	25.1	25.2	11.1	7.6	5.0	5.7	1.8
Haryana	10.8	18.6	14.3	8.5	1.9	3.6	3.0	1.4
Karnataka	33.1	42.5	37.7	27.0	8.3	10.2	10.2	6.4
Kerala	46.3	42.1	35.9	22.2	12.8	11.8	9.8	4.6
Madhya Pradesh	47.5	48.2	47.4	31.6	12.5	12.7	13.1	7.2
Maharashtra	56.3	40.6	48.8	28.3	15.9	10.0	13.4	5.9
Orissa	50.8	52.6	46.1	35.7	15.0	14.5	12.9	9.4
Punjab	11.2	14.4	7.7	6.2	2.0	2.8	1.9	0.9
Rajasthan	24.9	22.8	20.3	18.0	5.6	4.8	5.0	4.1
Tamil Nadu	43.1	39.8	43.5	39.1	10.8	10.0	11.4	10.5
Uttar Pradesh	37.9	41.2	32.0	31.1	9.3	8.7	7.1	7.1
West Bengal	61.1	62.5	56.5	48.6	20.3	20.9	17.0	14.1
All India	42.1	41.3	37.9	30.3	11.3	10.9	9.9	7.3

change for Madhya Pradesh, Rajasthan and Tamil Nadu, and positive (poverty incidence decreased) in the states of Bihar, Haryana, Karnataka, Orissa, Punjab, Uttar Pradesh and West Bengal. It is interesting to observe that the two richest states, viz, Punjab and Haryana registered a substantial increase in poverty between the first two years.

During the (1973-74 to 1977-78) period the head-count ratio for the poor was reduced in 13 states; the 2 states where it increased are Assam (3.1 per cent) and Maharashtra (1.9 per cent). The head-count ratio for the ultra poor was reduced only in 11 states, Gujarat and Tamil Nadu were the 2 states where the head-count ratio for the poor decreased whereas that for the ultra poor increased.

Between 1977-78 and 83, poverty reduced in all states except in Bihar—where it increased slightly. Thus, the decline in poverty was more rapid and widespread during this period compared to that in the previous one. The same conclusion seems to emerge whatever poverty measure is used. Also, sustained and significant decline over the two periods could be noticed only for Andhra Pradesh, Haryana, Kerala, Orissa, Punjab and West Bengal.

Table 9 gives the ranking of states by head-count ratio for both poor and ultra poor.

TABLE 9: RANKING OF STATES BY HEAD COUNT RATIO: POOR AND ULTRA POOR

	Poor				Ultra Poor			
	1972-73	1973-74	1977-78	1983	1972-73	1973-74	1977-78	1983
Andhra Pradesh	10	11	12	12	9	10	12	12
Assam	8	10	6	8	12	11	10	11
Bihar	4	2	2	1	4	3	3	1
Gujarat	11	12	13	13	10	12	11	14
Haryana	14	14	14	14	15	14	14	14
Karnataka	12	9	8	9	11	5	7	8
Kerala	6	7	10	10	6	6	8	9
Madhya Pradesh	5	4	4	5	5	4	4	5
Maharashtra	2	5	3	7	2	8	2	7
Orissa	3	3	5	4	3	2	5	4
Punjab	15	15	15	15	14	15	15	15
Rajasthan	13	13	13	11	13	13	13	10
Tamil Nadu	7	6	7	3	7	9	6	3
Uttar Pradesh	9	8	9	6	8	7	9	6
West Bengal	1	1	1	2	1	1	1	2

TABLE 10: GROWTH AND INEQUALITY EFFECTS ON HEAD-COUNT RATIO (POOR)

States	1973-74 to 1977-78			1977-78 to 1983		
	Growth	Inequality	Total	Growth	Inequality	Total
Andhra Pradesh	-5.2	0.1	-5.1	-6.3	-0.2	-6.5
Assam	2.5	0.6	3.1	-7.0	0.8	-6.2
Bihar	-0.4	-0.2	-0.6	0.3	0.0	0.3
Gujarat	-5.9	3.5	-2.4	-5.2	-2.4	-7.6
Haryana	-1.7	-3.0	-4.7	-4.5	-2.6	-7.1
Karnataka	-3.5	0.9	-2.6	-3.2	-1.0	-4.2
Kerala	-5.5	1.5	-4.0	-3.9	-0.8	-4.7
Madhya Pradesh	-1.7	1.4	-0.3	-2.8	-1.1	-3.9
Maharashtra	-10.8	12.0	-2.0	-5.6	-0.2	-5.8
Orissa	-2.8	0.6	-2.2	-1.2	-1.1	-2.3
Punjab	-16.2	-5.4	-10.8	-0.1	-3.3	-3.4
Rajasthan	-24.1	19.7	-4.4	-2.9	1.6	-1.3
Tamil Nadu	-1.4	1.1	-0.3	-1.3	-0.2	-1.5
Uttar Pradesh	-6.6	-3.2	-3.4	-0.2	-0.9	-1.1
West Bengal	-1.3	-0.1	-1.4	-1.2	-0.2	-1.4
All India	-4.2	2.4	-1.8	-2.2	-0.5	-2.7

The states have been ranked in the descending order of poverty. There has been a substantial change in the ranking. The states which have considerably improved their relative positions over the 1972-73 to 1983 period are Andhra Pradesh, Gujarat, Kerala and Maharashtra. It is interesting that these results corroborate well with the trends in nutritional status of the ultra poor [Subbarao, 1989], which showed an improvement in these 4 states during 1970-80. Again, while it is difficult to establish empirically the reasons for relatively improved performance in terms of reduction in poverty and malnutrition in these 4 states, it is worth mentioning that Andhra Pradesh and Gujarat's economic growth was impressive during this entire period, Maharashtra has had a significant intervention in the form of Employment Guarantee Scheme and Kerala had substantial social sector investments and an effective public distribution system. The relative positions of Bihar, Tamil Nadu and Uttar Pradesh have deteriorated—states which have had neither an impressive record of economic growth, nor substantial interventions providing effective social security. It is interesting to observe that the relative position of Assam is considerably higher on the basis of ultra-poverty than that on total

poverty. This discrepancy is mainly attributable to a very low degree of inequality in the predominantly tribal Assam which favours the ultra poor more than the poor.

Finally in this section we attempt to answer the question whether the rate of poverty reduction was greater or smaller in the poorer (richer) states. For this purpose we estimated the following regressions.

1973 to 1977-78:

$$\text{GHP} = -11.8 + 0.16\text{HP}, R^2 = 0.41 \quad (0.05)$$

$$\text{GHUP} = -7.2 + 0.12\text{HUP}, R^2 = 0.13 \quad (0.09)$$

1977-78 to 1983:

$$\text{GHP} = -6.4 + 0.05\text{HP}, R^2 = 0.10 \quad (0.04)$$

$$\text{GHUP} = -8.4 + 0.07\text{HUP}, R^2 = 0.07 \quad (0.08)$$

where

HP = Head-count ratio for the poor

HUP = Head-count ratio for the ultra poor

GHP = Growth rate of the head-count ratio (poor)

GHUP = Growth rate of the head-count ratio (ultra poor)

The relationship between HP and GHP is significant at the five per cent level in the first period (1973-74 to 1977-78) but it becomes insignificant in the second period (1977-78 to 1983). It means that in the first period, the rate of reduction in poverty tended to be smaller in the poorer states. However, this relationship does not hold for the ultra poor. In the second period, the relationship between the growth in poverty and the level of poverty was found to be insignificant for both the poor and the ultra poor.

In the following section, we separately examine the effect of growth and inequality of consumption on poverty across the states, and analyse their changing roles in poverty reduction over time.

V

Impact of Economic Growth and Inequality on Poverty

Suppose θ is a poverty index which will be a function of 3 factors: (1) poverty line income z , (2) the mean per capita consumption of the society, μ , and, (3) consumption or income inequality. Inequality can be measured by a single inequality index (many of which are available in the literature), but more generally it should be represented by the parameters of the Lorenz curve. We assume that the poverty line z is fixed, then we can write

$$d\theta = \frac{\partial \theta}{\partial \mu} d\mu + \sum_{i=1}^k \frac{\partial \theta}{\partial m_i} dm_i \quad (4)$$

where $m_1, m_2, \dots, m_k$ are the parameters of the Lorenz curve. This equation allows us to decompose the change in poverty into two components: (1) the impact of growth on poverty when the distribution of income does not change, and (2) the effect of income redistribution when the total income of the

society remains unchanged.

It is obvious that if the growth rate of consumption per capita is positive, the first component in equation 4 will always be negative. The second component can be both negative and positive. If there is any 'trickle-down' mechanism at work, we should expect the second component in equation 4 to be either zero or negative. It is worth mentioning that even if the inequality measured by the Gini index or some other measure has increased during the observation period, the second component in equation 4 can still be either zero or negative. In this section we will provide quantitative estimates of each of the two components of equation 4 which would enable us to assess the extent to which the 'trickle-down' mechanism is operating in each of the 15 states as well as at the India level.

It is perfectly legitimate to argue that high initial inequality levels may act as a constraint on economic growth. Our analysis does not answer this question of the effect of inequality of growth. In this paper, we are basically concerned with the ex post assessment of the impact of whatever growth has taken place on poverty reduction in different states. Notwithstanding the limitation of our inability to capture causality in its entirety, our analysis nevertheless strongly complements previous attempts [e.g., Ahluwalia 1978], which focused exclusively on regressing poverty on agricultural output per capita at the all India level.

We adopted the following procedure to compute the components of equation 4.

Let H_1 and H_2 be the head-count ratios for the first and the second periods, respectively; the percentage change in poverty between the two periods will be $(H_2 - H_1)/H_1$. To compute the growth effect on poverty, we assume that the Lorenz curve in the second period is the same as in the first period, but the mean consumption has changed. Suppose $L_1(p)$ is the Lorenz curve in the first period and μ_2 the real mean consumption in the second period, then H_2^* given by $L_1(H_2^*) = \frac{z}{\mu_2}$

will be the head-count ratio in the second period under the assumption that the inequality in the distribution remained the same in the two periods. Thus, the growth effect on poverty will be given by H_2^*/H_1 , which will be negative (positive) if μ_2 is greater (less) than μ_1 (the mean consumption in the first period). The effect of income redistribution on poverty will then be given by the difference between the total percentage change in poverty, viz, $(H_2 - H_1)/H_1$ and the growth effect, as measured by $(H_2^* - H_1)/H_1$. Given H_1 , H_2 and H_2^* , one can compute the corresponding values of any poverty measure, which would in turn provide the estimates of the growth and the inequality effects for that poverty measure.

The numerical estimates of the growth and inequality effects on head-count ratio are presented in Tables 10 and 11 for the poor and the ultra poor, respectively. Table 12 shows the estimates of these effects on the

poverty gap ratio for the poor.

At the all India level total poverty declined at an annual rate of 1.8 per cent between 1973-74 and 1977-78. If the inequality as measured by the Lorenz curve had not changed between the two periods, the poverty would have declined at an annual rate of 4.2 per cent. Thus, the change in the relative distribution which occurred between 1973-74 and 1977-78 resulted in an increase in poverty at an annual rate of 2.4 per cent. A less than proportionate share of the benefits of high growth rates during this period trickled down to the poor.

In the period between 1977-78 and 1983, total poverty declined at an annual rate of 2.7 per cent. This faster decline in poverty occurred despite a slightly lower rate in the per capita consumption observed in this period. However, the impact of growth on poverty continues to be high at -2.2 per cent. The change in the distribution contributed to a modest reduction in poverty at an annual rate of 0.5 per cent. Although the growth rate was low in the second period, the poor benefited proportionally more than the non-poor. In the absence of direct attacks on poverty aimed at raising the consumption levels of the poor, one could have

concluded that the trickle down effects of growth were more dominant in the second period. However, in view of these interventions, the favourable effects of inequality-reduction on poverty in this period could be attributed at least partly to these direct attacks on poverty.

The inequality effect was positive in almost all the states (exceptions being Haryana, Bihar and West Bengal), during the first period (1973-74 to 1977-78). The reduction in poverty which took place in most states was due to the high growth rates which more than compensated the adverse effects of increasing inequalities in this period. The states, where change in inequality had a high and unfavourable effect on poverty were Gujarat, Maharashtra, Punjab, Rajasthan and to some extent, Uttar Pradesh. However, this situation changed in the second period. In almost all the states, the poor benefited proportionally more than the non-poor. The states in which the poor benefited proportionally more than the non-poor in the second period are Haryana, Gujarat, Maharashtra, Rajasthan and Punjab. The effects of growth and inequality on poverty reduction was generally low in the states with low consumption per capita.

TABLE 11: GROWTH AND INEQUALITY EFFECTS ON HEAD-COUNT RATIO (ULTRA POOR)

States	1973-74 to 1977-78			1977-78 to 1983		
	Growth	Inequality	Total	Growth	Inequality	Total
Andhra Pradesh	-7.7	-2.0	-5.7	-8.5	-1.0	-9.5
Assam	5.8	-2.9	2.9	-11.9	2.9	-9.0
Bihar	-0.7	-1.1	-1.8	0.4	0.2	-0.6
Gujarat	-8.6	-8.7	0.1	-7.1	-6.7	-13.8
Haryana	-3.8	-2.6	-6.4	-6.5	-2.5	-9.0
Karnataka	-5.5	2.5	-3.0	-5.0	-0.9	-5.9
Kerala	-8.4	4.5	-3.9	-5.9	-2.5	-8.4
Madhya Pradesh	-2.8	2.4	-0.4	-4.4	-2.7	-7.1
Maharashtra	-15.8	20.5	4.7	-8.5	-0.9	-9.4
Orissa	-5.1	1.9	-3.2	-1.7	-2.8	-4.5
Punjab	-29.3	14.8	-14.5	0.0	-3.9	-3.9
Rajasthan	-29.9	27.0	-2.9	-3.6	1.4	-2.2
Tamil Nadu	-2.3	4.5	2.2	-2.0	0.1	-1.9
Uttar Pradesh	-12.3	6.2	-6.1	-0.4	-0.1	-0.5
West Bengal	-2.4	-0.1	-2.5	-2.1	-0.6	-2.7
All India	-6.2	4.2	-2.1	-3.2	-0.8	-4.0

TABLE 12: GROWTH AND INEQUALITY EFFECTS ON POVERTY-GAP RATIO (POOR)

States	1973-74 to 1977-78			1977-78 to 1983		
	Growth	Inequality	Total	Growth	Inequality	Total
Andhra Pradesh	-7.4	1.5	-5.9	-8.1	-1.4	-9.5
Assam	4.3	-2.6	1.7	-10.6	1.7	-8.9
Bihar	-0.7	-0.9	-1.6	0.5	-0.2	0.3
Gujarat	-8.8	9.2	0.4	-7.0	-6.7	-13.7
Haryana	-3.5	-2.2	-5.7	-6.1	-4.0	-10.1
Karnataka	-5.9	4.2	-1.7	-4.9	-1.4	-6.3
Kerala	-8.3	3.6	-4.7	-5.9	-3.3	-9.2
Madhya Pradesh	-3.3	3.3	0	-4.7	-2.5	-7.2
Maharashtra	-15.7	19.9	4.2	-9.0	-0.8	-9.8
Orissa	-5.6	2.0	-3.6	-1.8	-2.4	-4.2
Punjab	-17.2	6.5	-10.7	-0.4	-6.1	-6.5
Rajasthan	-29.8	27.5	-2.3	3.8	1.6	5.4
Tamil Nadu	-2.5	3.8	1.3	-2.1	0.5	-1.6
Uttar Pradesh	-11.8	8.1	-3.7	-0.4	-0.9	-1.3
West Bengal	-3.0	-0.4	-3.4	-2.4	-0.3	-2.7
All India	-6.1	4.3	-1.8	-1.7	-1.0	-2.7

VI

Responsiveness of Poverty to Growth and Inequality Changes

Even if growth is trickling down, because of initial high inequality levels, the effect of growth on poverty may be small. To see how growth affects poverty, we computed the poverty elasticities with respect to the mean per capita consumption when the effect of changes in inequality has been kept constant. These elasticities which will be referred to as growth elasticities have been derived by Kakwani [1989] for all the existing poverty indices. The elasticities for the three poverty measures used in the present paper are:

$$\begin{aligned}\text{Head-count ratio} &= \frac{zf(z)}{H} \\ \text{Poverty-gap ratio} &= \frac{\mu^*}{(z - \mu)^*} \\ \text{Watts' measure} &= \frac{H}{W}\end{aligned}$$

where W represents Watts' measure, H is the head-count ratio, and μ^* is the mean per capita consumption of the poor.

The measurement of the effect of inequality on poverty is a more difficult task because inequality in distribution can change in infinite ways. To get an idea of the size of this effect we make a simple assumption that the entire Lorenz curve shifts according to the following formula:

$$L^*(p) = L(p) - \lambda(p - L(p)) \quad (5)$$

which implies that when $\lambda > 0$, the Lorenz curve shifts downwards resulting in higher inequality. It can be shown that λ is equal to the proportional change in the Gini index. If $\lambda = 0.01$, it means that the Gini index is increased by 1 per cent. Thus, one can derive the elasticity of a poverty measure with respect to the Gini index using this procedure.⁶ The elasticities for the 3 poverty measures used in the present study are:

$$\begin{aligned}\text{Head-count ratio} &= \frac{(\mu - z)}{H} f(z) \\ \text{Poverty-gap ratio} &= \frac{(\mu - \mu^*)}{(z - \mu)^*} \\ \text{Watts' measure} &= \frac{H}{W} \frac{(\mu - h)}{h}\end{aligned}$$

where h is the harmonic mean of the income distribution of the poor only.

The inequality elasticities of the poverty-gap ratio and Watts measures will always be positive, i.e., the higher income inequality leads to greater poverty. For the head-count measure this result is not always true. If $\mu < z$, the head-count measure can decrease with an increase in inequality. This result casts down the usefulness of the head-count measure for analysing the impact of inequality on poverty.

Because the mean income and income inequality each affects poverty, an important question arises: What is the trade-off between mean income and income inequality? Put differently, we may ask, if the Gini index

of the income distribution increases by one per cent, what would be the percentage increase in the mean income for poverty not to increase at all? This can be answered with the concept of the marginal proportional rate of substitution (MPRS) between mean income and income inequality.⁷ It is given by:

$$\text{MPRS} = \frac{\left| \frac{\partial \mu}{\partial G} \right|}{\mu} = - \frac{\text{inequality elasticity of poverty}}{\text{growth elasticity of poverty}}$$

which can be computed for each poverty measure.

Tables 10 and 11 present the growth and inequality elasticities for the poverty-gap ratio for the poor and ultra poor, respectively. The values of MPRS for the poverty-gap ratio are presented in Table 12. The conclusions emerging from these tables are summarised below.

Both growth and income elasticities have a general tendency to increase over time, the increase being slower in the poorer states. Thus, the poverty ratio is becoming more responsive to the changes in the mean income and income inequality. In the two poorest states, viz, Bihar and West Bengal, the elasticities have not changed much sug-

gesting fairly stable distributions.

The inequality elasticity has increased slightly faster than the growth elasticity in most states. This pattern is quite evident from the increasing values of the marginal proportionate rate of substitution (MPRS) presented in Table 15. This suggests that a

greater emphasis should be placed on the growth oriented policies which at least maintain the income share of the poor. But if the inequality deteriorates during the course of economic growth, poverty may increase even with a faster economic growth because of increasing inequality elasticity.

Since the values of MPRS are generally higher for the richer states, it would appear imperative to follow consumption redistribution policies even in the high growing states.

Both growth and inequality elasticities are considerably higher for the ultra poor than for the poor implying that increasing inequality will hurt the ultra poor more than the poor. Also, the positive change in growth rate will benefit the ultra poor more than the poor. Since the MPRS values are considerably higher for the ultra poor than for the poor, the inequality reducing policies (in comparison with the growth policies) will have to play a more important role if the

TABLE 13: GROWTH AND INEQUALITY ELASTICITY ON POVERTY-GAP (POOR)

States	Growth Elasticity			Inequality Elasticity		
	1973-74	1977-78	1983	1973-74	1977-78	1983
Andhra Pradesh	-2.6	-2.8	-3.5	1.6	2.1	3.3
Assam	-3.2	-3.5	-4.3	1.2	0.9	1.7
Bihar	-1.9	-2.1	-2.0	0.7	0.7	0.7
Gujarat	-3.1	-2.7	-4.4	1.6	1.9	3.4
Haryana	-3.1	-3.3	-4.2	2.8	3.1	4.4
Karnataka	-2.2	-2.1	-2.5	1.2	1.5	2.1
Kerala	-2.0	-2.1	-3.0	1.2	1.8	2.8
Madhya Pradesh	-2.0	-2.0	-2.6	1.0	1.1	1.7
Maharashtra	-2.3	2.0	-2.8	1.0	1.8	1.8
Orissa	-1.8	-2.0	-2.4	0.7	1.0	1.1
Punjab	-3.5	-3.5	-4.4	3.2	4.8	5.6
Rajasthan	-3.0	-2.6	-2.8	2.2	4.4	3.4
Tamil Nadu	-2.3	-2.1	-2.1	1.0	1.2	1.4
Uttar Pradesh	-2.6	-2.6	-2.6	1.0	1.7	1.8
West Bengal	-1.5	-1.7	-1.9	0.5	0.7	0.8
All India	-2.21	-2.22	-2.48			

TABLE 14: GROWTH AND INEQUALITY ELASTICITY ON POVERTY-GAP (ULTRA POOR)

States	Growth Elasticity			Inequality Elasticity		
	1973-74	1977-78	1983	1973-74	1977-78	1983
Andhra Pradesh	-1.9	-3.3	-4.0	2.3	3.7	5.5
Assam	-3.6	-5.0	-5.6	2.4	2.5	3.9
Bihar	-2.5	-2.7	-2.8	1.4	1.6	1.4
Gujarat	-4.0	-3.4	-5.2	3.2	3.7	5.9
Haryana	-4.2	-3.8	-5.1	5.2	5.2	7.5
Karnataka	-3.2	-2.7	-3.2	2.2	2.6	3.6
Kerala	-2.6	-2.7	-3.8	2.2	3.1	5.0
Madhya Pradesh	-2.8	-2.6	-3.4	1.9	2.1	3.1
Maharashtra	-3.1	-2.6	-3.8	2.1	3.2	3.6
Orissa	-2.6	-2.6	-2.8	1.5	1.9	2.2
Punjab	-4.1	-3.1	-5.9	5.6	6.4	10.2
Rajasthan	-3.8	-3.1	-3.4	4.0	6.4	5.6
Tamil Nadu	-3.0	-2.8	-2.7	2.1	2.3	2.6
Uttar Pradesh	-3.7	-3.5	-3.4	2.2	3.3	3.1
West Bengal	-2.0	-2.3	-2.4	1.0	1.3	1.7
All India	-2.80	-2.82	-3.15			

ultra poor are to be lifted out of their desperate situation.

The policies pursued by the government since the mid-1970s with the objective of raising the consumption levels of the poor and the ultra poor, therefore, appear to be basically sound responses to the observed effects of growth and inequality analysed in this part. In the following sections the effectiveness of these strategies are evaluated at the state level.

VII

Poverty Alleviation Policies

The government's strategy for reducing the incidence of poverty is based on a combination of accelerated overall growth and targeted direct anti-poverty interventions. In the long run, the outlook for poverty reduction depends on: (a) the supply of, and the demand for, labour, and (b) on the social policies that help raise the basic capabilities of people, especially education and health. These aspects are analysed below.

(A) POLICIES FOR LABOUR ABSORPTION

Given that the entrants to the labour force are unlikely to register a downward trend over the next three decades, the critical issues are: (i) to what extent can future growth in agriculture and outside agriculture absorb the growing labour force? and (ii) to what extent can direct interventions help in expanding self-employment and wage employment especially among the ultra poor?

(i) Agricultural Growth and Poverty

Reference was already made to the empirical evidence establishing the close link between agricultural growth and poverty reduction. Rao [1987] noted that the rate of reduction in poverty incidence was lowest in agriculturally lagging regions. Table 16 gives the distribution of states classified by the differences in the rates of agricultural growth and the degree of instability of growth. As many as 70 per cent of the nation's rural poor as well as ultra poor are located geographically in states with low or moderate rates of agricultural growth with medium or high instability. The regions experiencing

high or medium growth with low instability account for only 27 per cent of the poor and 25 per cent of the ultra poor.

Given the importance of agricultural progress in poverty reduction, the important questions are: What are the prospects of growth in the lagging regions? Would an accelerated growth in these regions lead to adequate labour absorption?

Recent evidence suggests a fall in aggregate employment elasticity of agriculture over the 1970s [Bhalla, 1987] but this is a composite estimate reflecting the experience of diverse regions and crops. Employment elasticities with respect to output continue

TABLE 15 MARGINAL PROPORTION RATE OF SUBSTITUTION BETWEEN GROWTH AND INEQUALITY POVERTY-GAP RATIO

States	Poor			Ultra Poor		
	Growth	Inequality	Total	Growth	Inequality	Total
Andhra Pradesh	0.6	0.8	0.9	1.2	1.1	1.4
Assam	0.4	0.3	0.4	0.7	0.5	0.7
Bihar	0.4	0.3	0.4	0.6	0.6	0.5
Gujarat	0.5	0.7	0.8	0.8	1.1	1.1
Haryana	0.9	0.9	1.0	1.2	1.4	1.5
Karnataka	0.5	0.7	0.8	0.7	1.0	1.1
Kerala	0.6	0.9	0.9	0.8	1.1	1.3
Madhya Pradesh	0.5	0.6	0.7	0.7	0.8	0.9
Maharashtra	0.4	0.9	0.6	0.7	1.2	0.9
Orissa	0.4	0.5	0.5	0.6	0.7	0.8
Punjab	0.9	1.4	1.3	1.4	2.1	1.7
Rajasthan	0.7	1.7	1.2	1.1	2.2	1.6
Tamil Nadu	0.4	0.6	0.7	0.7	0.8	1.0
Uttar Pradesh	0.4	0.7	0.7	0.6	0.9	0.9
West Bengal	0.3	0.4	0.4	0.5	0.6	0.7

TABLE 16: DISTRIBUTION OF POOR ACCORDING TO LEVELS OF GROWTH AND INSTABILITY IN FOODGRAINS PRODUCTION

Category	States	Per Cent of New Sown Area with Rainfall			Per Cent in			Number of Agricultural Workers Per 100 Hectares of Net Sown Area, 1980-83
		High (1,150 mm and above)	Medium (750 mm to 1,150 mm)	Low (up to 750 mm)	Rural Population, 1983	Poverty Population, 1983	Ultra Poor	
1	2	3	4	5	6	7	8	9
I High growth and low instability	Punjab, Jammu and Kashmir	5.3	20.5	74.2	3.2	0.9	NA	85
II High growth and medium instability	Haryana, Uttar Pradesh	8.4	64.1	27.5	19.6	18.7	19.0	148
III High growth and high instability	Gujarat	7.3	25.0	67.7	4.5	2.0	2.73	83
IV Medium growth and low instability	Andhra Pradesh, Himachal Pradesh	1.0	66.0	33.0	8.7	4.5	3.9	145
V Medium growth and medium instability	Karnataka	9.4	24.3	66.3	5.1	4.4	4.6	95
VI Medium growth and high instability	Maharashtra, Orissa, Rajasthan	25.0	24.1	50.9	17.6	17.6	16.7	87
VII Low growth and low instability	Assam, Kerala	100.0	Nil	Nil	7.5	5.4	6.5	173
VIII Low growth and medium instability	Tamil Nadu, West Bengal	54.2	45.8	Nil	14.0	19.6	18.4	188
IX low growth and high instability	Bihar, Madhya Pradesh	63.5	33.5	3.0	19.8	26.9	26.1	130
	All India	30.7	35.8	33.5	100.0	100.0		120

Note: Growth rate refers to the foodgrains production growth rate for the period 1961-1985: High-above 3.0; Medium - 2.0 to 3.0; Low-below 2.0. Instability refers to the standard deviation in the annual output growth rates of foodgrains for the period 1961-1985; High-above 20; Medium-15 to 20; Low-below 15.

Source: All columns except 8, C H H Rao, S K Ray, and K Subbarao [1988]; column 8 was computed in this paper.

to be higher in the low-wage eastern states for all crops than in the rest of India. Diffusion of agricultural technology to the low-wage regions is bound to result in greater labour absorption. This may be already happening in eastern Uttar Pradesh with the emergence of HYV rice as a major second crop, although one may have to wait for years for reliable empirical evidence.

Accelerating agricultural growth in the eastern region, a region where our results have shown the smallest reduction in poverty, calls for a significant reorientation of science and technology policies as well as institutional policies so as to help strengthen the small farm sector. In particular, science and technology policies have to focus on the development of high-yielding *crop mixes* that facilitate intercropping so that total output from a crop rotation is maximised. If the country is to take advantage of the reser-

voir of labour available in the small farm sector in this region, this sector needs to be strengthened both by technology and infrastructure policies that promote higher cropping intensities, and by institutional policies that enhance their resource base, especially consolidation of holdings, and credit and marketing reforms [C H H Rao, S K Ray, K Subbarao, 1989]. Progress on all these fronts has been far from satisfactory.

Rural non-farm activities—dairying, poultry, fisheries, forestry and agro-processing—have been a major source of poverty redress in India. These activities are now being encouraged through Integrated Rural Development which is discussed in later sections.

(ii) Non-Agricultural Growth and Poverty

In the long run, agriculture's share in employment is bound to fall, so that growth

in manufacturing and service sector employment would be crucial in the coming decades. Past experience is very disappointing. Capital intensity in manufacturing was maintained by a policy environment that favoured existing workers and jobs. Labour demand elasticity with respect to the wage for the period 1973-74 to 1984-85 ranged between -0.7 and -0.8, implying a significant trade-off between real wages and industrial employment [Hanson and Sengupta, 1989]. There has been a rise in the real wage of organised labour in both public and private sectors in the 1980s contributing to slow growth of employment in manufacturing. Prospects for non-farm employment depend on (a) fiscal and other measures to stimulate light labour-intensive industry including export industry, and (b) public policy to stimulate service sector (including rural informal sector). As of now, there has been no specific policy framework that encourages labour-intensive manufacturing and service and informal sectors.

(iii) Direct Anti-Poverty Interventions

Since the mid-1970s, the central and the state governments have launched numerous direct anti-poverty interventions whose scale and variety is so vast that it is not possible to review them all in this paper. The important direct interventions launched with the initiative of the central government are: Integrated Rural Development Programme (IRDP), National Rural Employment Programme (NREP), and the Rural Landless Employment Guarantee Programme (RLEGP). There is also an important centrally-sponsored programme for combating child and maternal malnutrition, i.e. the Integrated Child Development Services (ICDS). At the state level, the notable programmes are: Maharashtra's Employment Guarantee Scheme (MEGS); public distribution systems in Kerala, Tamil Nadu, Gujarat, and Andhra Pradesh (Rs 2-a-kilo scheme); and Tamil Nadu Integrated Nutrition Project. With regard to each of these programmes, there have been many evaluations and much debate. In view of our finding that in the second period, despite slower growth in average consumption per capita, the poverty ratio fell largely due to a decline in inequality in most states, an overall assessment of anti-poverty programmes initiated during this period is necessary. Any such assessment of their overall impact, however, must begin with the recognition of the fact that India is experimenting in many directions to reduce poverty-induced human suffering.

IRDP has been in operation long enough to be realistically evaluated. Independent analysts recognise that this programme has channelled funds on a hitherto unprecedented scale for creating supplementary incomes from non-land assets among the relatively poor all over rural India. During the Sixth Plan period, assets worth some Rs 50 billion were created and distributed benefiting about 17 million families—by no means a small achievement. During 1987-88,

TABLE 17: MAJOR POVERTY ALLEVIATION PROGRAMMES, 1987-88

A Financial Expenditures	Rs Million
1 Credit-based self-employment programmes:	
a Integrated Rural Development Programme (IRDP)*	19,000
(Investment per beneficiary = Rs 4.471)	
2 Wage employment programmes:	
a National Rural Employment Programme (NREP)	7,850
b Rural Landless Employment Guarantee Programme (RLEGP)	6,480
c Maharashtra's Employment Guarantee Scheme (MEGS)	2,650
3 Area development programme	900
a Drought Prone Area Programme	900
b Desert Development	500
c Watershed Development	2,400
	39,780
As per cent of GDP	1 per cent
As per cent of plan outlay	9 per cent
B Physical Achievements, 1987-88	
1 Credit-based self-employment programmes:	1987-88
a IRDP: Number of beneficiaries covered (million)	4.2
2 Wage employment programmes:	
a NREP: Mandays of employment generated (million)	370
b RLEGP: Mandays of employment generated (million)	100
c MEGS: Mandays of employment (million)	150

Source: Progress Reports of the 20 Point Programme.
Ministry of Programme Evaluation.

TABLE 18: STATE-WISE DISTRIBUTION OF IRDP FUNDS AND POOR

States	Percentage Distribution of		
	IRDP Expenditure, 1988-89	Poor Population (1983)	Ultra Poor Population (1983)
Andhra Pradesh	8.6	5.01	3.9
Assam	1.7	3.24	2.1
Bihar	12.8	17.44	20.0
Gujarat	3.0	2.73	1.7
Haryana	1.6	0.86	0.6
Karnataka	4.5	4.56	4.6
Kerala	2.6	3.30	3.0
Madhya Pradesh	9.7	8.66	8.5
Maharashtra	8.0	7.95	7.5
Orissa	4.0	5.27	5.4
Punjab	1.6	0.77	0.5
Rajasthan	4.2	3.53	3.2
Tamil Nadu	7.2	7.32	8.2
Uttar Pradesh	21.3	18.23	18.3
West Bengal	9.0	11.12	12.7
All India	100	100	100

the fourth year of the Seventh Plan, another 4.2 million families were assisted with an investment of Rs 4,471 per family, or Rs 19 billion overall (see Table 17). So far, IRDP reached about 25 per cent of India's rural households.

State-wise shares of IRDP expenditures are compared with their shares in the population of the poor and the ultra poor in Table 18. On the basis of the incidence of either poor or ultra poor population, the eastern states of Assam, Bihar and West Bengal deserved higher allocations.

There are numerous evaluations of IRDP—both by government departments and private researches. In the past, most assessments by the government have been favourable, but micro studies of IRDP have not been unequivocal in supporting the official claims. However, the divergence between macro indicators and micro performance seem to be narrowing in recent years. Also, 'success' or 'failure' depend on the criterion adopted for assessment. Thus, Subbarao [1985] argued that the most stringent criterion of "crossing the poverty line" is inappropriate for judging the full benefits of this programme, since households way below the poverty line (the ultra poor) may register incremental incomes and thus benefit from the programme, even if they are unable to cross the poverty-income threshold.⁸ Pulley [1989] has shown that depending upon the criterion adopted, programme success rate varied (see Table 19). On the basis of the criterion of "investments remaining intact" (which suggests that households are deriving incremental incomes from the asset), the programme is doing reasonably well even in relatively low income states like Bihar. But on the basis of the rigorous criterion of "crossing the poverty line", the success rate was very low (see Table 19, columns 3-5). State-wise performance also suggests in interesting relationship, between "percentage of eligible beneficiaries", and the proportion crossing the poverty line. Barring in the hill states (Himachal Pradesh, Jammu and Kashmir), the higher the proportion of eligible beneficiaries (i.e., the lower the percentage of non-poor) with low initial level of incomes, the more difficult it is for them to cross the poverty income threshold; the proportion doing so is small.

One of the major criticisms of IRDP is that the programme is benefiting the households closer to the poverty line income threshold, and the ultra poor are unable to take advantage of the scheme. It is suggested that the poorest households may not be able to hold and manage assets. Recent evidence, however, effectively refutes this criticism. In fact, in an administratively weak and relatively poorer state like Uttar Pradesh, it has been found, based on panel data for 4 years, that the poorest households not only managed to hold on to assets, but also found to be deriving income from the assets *on a sustained basis* [C H H Rao and Rangaswamy; and Pulley]. The problem encountered by

these households was the continued reluctance of the institutional credit agencies to lend working capital on a regular basis *even after the households have proven their credit-worthiness* by promptly repaying the IRDP loans. In other words, IRDP enabled access to institutional credit for the poorest households as a *one-shot* injection, but failed to open a channel of line credit on a continuing basis for the neediest households notwithstanding their proven creditworthiness.

In brief, recent evidence suggests that the very order of investments in the IRDP is such that it might be able to give impetus

for the development of supportive infrastructure and raise the incomes of the poor at the margin, provided the choice of assets are properly dovetailed with the level and structure of demand in the region; the assets supplied were labour-intensive in character, and supporting marketing networks were simultaneously promoted. Wherever these conditions were satisfied, and the programme reached the lower half of the poverty groups, its impact on the hard-core poor was not insubstantial. However, for *sustained* income generation, it would be necessary to ensure continued access to institutional

TABLE 19: MEASURES OF SUCCESS OF IRDP

Major States	Per Cent Eligible* Beneficiaries	Per Cent Investments Intact**	Per Cent Intact and No Credit Overdue	Per Cent Eligible and Crossed Poverty Lines***
1	2	3	4	5
Andhra Pradesh	68	76	34	9
Arunachal Pradesh	73	61	38	4
Assam	27	70	6	10
Bihar	76	85	18	3
Gujarat	78	88	43	4
Haryana	71	46	15	0
Himachal Pradesh	87	85	45	29
Jammu and Kashmir	97	80	50	19
Karnataka	85	64	26	4
Kerala	89	74	19	5
Madhya Pradesh	81	73	27	6
Maharashtra	83	69	30	10
Orissa	83	68	19	7
Punjab	30	77	57	18
Rajasthan	72	48	15	9
Tamil Nadu	83	63	28	3
Uttar Pradesh	54	79	41	5
West Bengal	46	97	23	8
Average	70	73	29	7

Note: * Proportion of beneficiaries with pre-IRDP income $\leq$ 4800.

** Proportion of IRDP investments that remained fully operational after two years.

*** Proportion of beneficiaries with pre-IRDP income $\leq$ 4800 and post-IRDP income = 6400 after two years in current price terms.

Source: National Concurrent Evaluation of IRDP, Ministry of Rural Development, as quoted in Pulley [1989].

TABLE 20: NREP—STATE-WISE DISTRIBUTION OF EMPLOYMENT

	Employment (Million Mandays)	Per Cent Women Employed	Percent Distribution of Employment	
				Ultra Poor Population
Andhra Pradesh	34.0	41.8	9.0	3.9
Assam	2.7	Nil	0.7	2.1
Bihar	48.0	10.0	12.8	20.0
Gujarat	13.3	28.6	3.5	1.7
Haryana	2.5	Nil	0.7	0.6
Karnataka	21.7	13.4	5.8	4.6
Kerala	15.0	26.7	4.0	3.0
Madhya Pradesh	31.4	27.1	8.4	8.5
Maharashtra	25.9	26.2	6.9	7.5
Orissa	18.3	25.7	4.9	5.4
Punjab	2.9	Nil	0.8	0.5
Rajasthan	22.7	24.2	6.0	3.2
Tamil Nadu	44.2	44.1	11.8	8.2
Uttar Pradesh	81.3	3.1	21.6	18.3
West Bengal	11.8	3.0	3.1	12.7
All India	379.6	20.65	100.00	100.0

Source: Ministry of Programme Implementation Progress Reports.

credit for the very poor.

The overall assessment of the two employment programmes (NREP and RLEGP) is similar. Together these programmes provided, on average, about 450 million mandays of employment per annum. Here again, evaluations pointed out that: (a) the impact of the programmes on the total income of

poor households was insignificant because they met only about 9 per cent of the demand for work from the poor in rural areas; (b) the programmes could not create sufficiently useful, wage-intensive works at times and in places most needed; (c) the poor could not benefit from the assets created; (d) the assets created were of poor quality; and (e) wages actually paid were lower than budgeted owing to leakages and corruption. It must be stressed that some of these criticisms are misplaced. For example, the size of the programme (NREP) is not small, when compared to *incremental* employment generated in the rural areas. One recent estimate [Subbarao, 1987a] suggests that NREP provided nearly 40 per cent of the total *incremental* employment in rural India.

There is, however, no doubt that MEGS, a state-level programme, has a much better record, both in generating supplementary employment especially for women, and also

in terms of much better administered wage structure that reduced gender differences in wage rates. Nonetheless, recent moves to pay statutorily fixed minimum wages may destroy some of the merits of this scheme [Subbarao, 1987a] such as its self-targeting character.

Only two states—Andhra Pradesh and

Tamil Nadu—have been successful in attracting women in large numbers to NREP work sites (see Table 20). The state with a large proportion of the ultra poor—Uttar Pradesh—has a dismal record in the employment of women in NREP projects. However, the state-wise distribution of mandays of employment generated does appear to be in

TABLE 22: PER CAPITA PUBLIC EXPENDITURE (REVENUE)^a ON HEALTH BY STATES

	Per Capita Expenditure at 1980-81 Prices		Rate of Growth	Life Expectancy		Infant Mortality	
	1976-77	1986-87		1976	1986	1976	1985
Andhra Pradesh	18.1	30.4	5.3	47.9	53.1	122	83
Bihar	8.7	15.0	5.6	42.3	46.0	NA	105
Gujarat	20.2	39.6	7.0	50.2	52.4	146	98
Haryana	20.4	37.5	6.3	52.9	54.8	112	85
Karnataka	20.0	23.2	1.5	54.5	56.3	89	71
Kerala	24.8	29.3	1.7	61.7	65.5	56	32
Madhya Pradesh	17.6	18.3	0.4	46.9	49.0	138	122
Maharashtra	20.3	44.7	8.2	53.5	56.3	83	68
Orissa	15.7	32.2	7.4	44.0	49.1	127	130
Punjab	24.1	32.8	3.1	58.4	60.5	108	71
Rajasthan	21.4	32.8	4.4	49.3	51.9	142	108
Tamil Nadu	22.9	33.3	3.8	50.3*	53.4	110	80
Uttar Pradesh	10.0	19.1	6.7	42.8	46.2	178	140
West Bengal	22.0	25.4	1.4	49.6	52.0	NA	77

Source: RBI Bulletins, Various Issues.

a Includes expenditure on family welfare and water supply.

TABLE 21: REGIONAL PROFILE OF MALNUTRITION AMONG CHILDREN (PERCENTAGE OF SEVERELY MALNOURISHED CHILDREN (GRADES III AND IV) IN ICDS PROJECT AREAS DURING 1984-85^a

Sl No	State	District	Type of Project ^b	Per Cent Malnourished		Child Mortality Rate ^c		District Characteristics		
				0-3	3-6	0-2	0-5	SC + ST ^d	AVA POP ^e	IRR ^f
				(1)	(2)	(3)	(4)	(5)	(6)	(7)
1	Andhra Pradesh	Guntur	Rural	8.80	11.30	99	142.00	13.00	317.00	39.00
		Vizianagaram	Tribal	22.20	32.50	169	196.00	19.00	61.00	30.00
2	Karnataka	Dharwar	Rural	15.40	12.90	124	182.00	15.00	307.00	6.00
		Mysore	Rural	17.20	10.80	109	149.00	24.00	270.00	17.00
3	Tamil Nadu	Kanyakumari	Urban	5.70	4.60	70	80.00	5.00	161.00	52.00
		Salem	Urban	5.70	0.30	74	101.00	20.00	245.00	22.00
4	Kerala	Mallapuram	Rural	1.20	4.30	66	104.00	9.00	355.00	57.00
		Palghat	Rural	0.70	4.00	74	116.00	19.00	421.00	57.00
5	Madhya Pradesh	Jabalpur	Rural	34.40	16.90	220	265.00	29.00	152.00	5.00
		Dhar	Tribal	26.00	17.30	138	174.00	59.00	350.00	7.00
6	Uttar Pradesh	Pratapgarh	Rural	14.60	13.00	163	211.00	22.00	222.00	28.00
		Kheri	Rural	6.10	3.10	154	187.00	27.00	587.00	10.00
7	Rajasthan	Bharatpur	Rural	12.40	na	214	252.00	24.00	369.00	20.00
		Banswara	Tribal	14.20	0.70	170	169.00	25.00	235.00	7.00
8	Maharashtra	Yaystmal	Rural	22.30	25.10	177	227.00	26.00	264.00	2.00
		Nanded	Urban	12.00	10.20	87	107.00	21.00	289.00	3.00
9	Gujarat	Valsad	Tribal	4.60	3.30	91	96.00	57.00	192.00	13.00
		Baroda	Tribal	7.30	5.60	115	129.00	31.00	206.00	21.00
10	Orissa	Cuttack	Rural	3.20	2.80	196	204.00	21.00	271.00	32.00
11	West Bengal	Nadia	Rural	21.30	12.20	115	149.00	28.00	268.00	33.00
		Bankura	Tribal	6.50	8.20	77	89.00	40.00	427.00	33.00
12	Haryana	Hissar	Rural	6.70	4.20	120	146.00	22.00	901.00	64.00
		Bhiwani	Rural	5.50	0.70	114	129.00	18.00	558.00	20.00

Notes: a = The data in columns 1 and 2 are taken from ICDS Project-wise Progress Report for 1984-85, for those projects which were sanctioned during 1983-84. As such, these figures reflect the nutrition situation at/about the commencement of ICDS in these districts.

b = rural; T = tribal; U = urban.

c = District-wise child mortality rates for 1981 for the age groups less than 2 and less than 5 years were made available for the first time in the Occasional Paper 5, 1988, Office of Registrar-General, Ministry for Income Affairs.

d = Per cent SC/ST to total population.

e = Agricultural value-added.

f = Irrigation ratio.

conformity with the distribution of ultra poor. It would be interesting to see whether the state-wise distribution of mandays of employment generated is correlated with the distribution of the ultra poor. We estimated this correlation which was found to be 0.78 statistically significant. However, there are some outliers. For example in Bihar and West Bengal, employment generated was substantially lower than the share of the ultra poor.

Most evaluations of employment have expressed concern that states have opted for rural roads, primary school building construction, etc, in preference to directly productive activities such as soil conservation and watershed development. Recent work by Binswanger, et al [1989] points to the significant positive impact of rural infrastructure (markets and roads) on agricultural output. Therefore, there appears to be no strong argument for discouraging rural infrastructure building on grounds that they are "not directly productive". The most immediate concern ought to be ensuring quality of works and maintenance of infrastructure created.

Despite the deficiencies noted in employment programmes, these schemes appear to have certain desirable features, such as a certain amount of self-targeting (through relative unattractiveness of this employment) and that the guarantee schemes can in a sense substitute for a social security system, at least for those who are able to work. If attempts are made to remove the observed deficiencies and women are attracted to work sites as in MEGS, these programmes do have the potential to reach the bottom half of the population within the poverty group.

(B) NUTRITION, HEALTH, AND OTHER SOCIAL SECTOR INTERVENTIONS

The basic objective of direct nutrition interventions is to protect *specific vulnerable groups*, such as children and women, from malnutrition. Food, being the basic need, direct intervention to increase food security for the poor may have good pay-off in terms of human welfare. An important nationwide programme is Integrated Child Development Services (ICDS) launched in 1975, whose aim is to provide a package of services such as child protection and development with stress on both pre-natal services including immunisation, periodical health check up, and referral services. Growth monitoring and supplementary nutrition for 300 days of the year for all children are the other components of the scheme. About 6.5 million children below 6 years of age and 1.5 million women were covered under ICDS during 1985-86.

The survey data of the National Nutrition Monitoring Bureau (NNMB) do show steady but slow decrease in severe malnutrition during 1974-81 in 10 states [Subbarao, 1989]. There has been no NNMB survey since 1981. ICDS baseline data are useful in assessing the recent geographical patterns of malnutri-

tion. These are shown in Table 21. In order to test that reliability of the number of the percentage of severely malnourished children collected by ICDS authorities, we estimated the correlation between these and the child mortality rates for the corresponding 23 districts. A statistically significant positive correlation between the two series suggests that the ICDS baseline data can be relied upon for assessing geographical variations in malnutrition at a fairly disaggregated level.

What can we infer from these data? The percentage of children severely malnourished are indeed very high in *tribal* Andhra Pradesh, Bihar, Rajasthan and Madhya Pradesh and also in *rural* Karnataka, Madhya Pradesh, Uttar Pradesh, Maharashtra and West Bengal. Considering that these states account for a significant proportion of the nation's ultra poor, and bearing in mind the concentration of poverty among the scheduled castes and tribes, *poverty-induced* malnutrition continues to be rampant in pockets of India.

As with other interventions, evaluations of ICDS have pointed out many shortcomings [Subbarao, 1989]. Immunisation levels continued to be low and there was a drop from the first to the second dose in many cases, the regularity of feeding differed from centre to centre, and there was little community participation. The programme, like most other interventions, has been relatively more successful in reaching the poor and poorer groups, but not the poorest. There are significant inter-project variations in

impact (Heaver), with relatively poorer states/regions faring worse than others owing especially to the inability of poor states to put in their share of expenditure on nutrition supplements. Regions with low infrastructural levels also could not do well [Subbarao, 1989]. ICDS is very promising and could become a good complement to other poverty alleviation programmes provided children of the poorest groups are attracted to the scheme, if necessary, by appropriate incentives such as provision of clothes to children. However, India's experience suggests the need for great caution in exercising choice among competing alternative strategies for filling nutrition needs of the poor. For example, in Tamil Nadu, which has a long experience in nutrition management, a package of nutrition and health services to children of 6 to 36 months old is being delivered under the Tamil Nadu Integrated Nutrition Project. This project was able to elicit community participation to a far greater extent than ICDS. Important non-economic factors contributed to its success such as recruitment of nutrition workers from among the deprived communities in which malnutrition was most severe [Subbarao, 1989].

HEALTH

Access to primary health care has a direct effect on the well-being of the poor in rural India. There is ample empirical evidence to suggest that India is much behind other countries with comparable per capita incomes in South Asia with respect to social

TABLE 23: URBAN AND RURAL LITERACY RATES, BY AGE AND SEX, 1981

Location/Sex	Age Groups			All Ages
	10-15	15-35	35 and Over	
(a) Literacy rates				
Urban				
Male	82.6	81.5	69.4	65.8
Female	72.7	63.6	35.9	47.8
Both	78.1	73.1	54.3	57.4
Rural				
Male	62.8	56.1	38.0	40.8
Female	36.0	25.7	8.6	18.0
Both	50.2	41.5	23.2	29.7
Total				
Male	67.2	63.9	44.6	46.9
Female	44.4	35.3	14.4	24.8
Both	56.5	50.0	30.2	36.2
(b) Percentage rise, 1971-81				
Urban				
Male	1.8	4.1	7.6	7.5
Female	3.7	13.8	18.9	13.5
Both	2.9	7.3	10.1	9.5
Rural				
Male	14.0	15.0	21.8	20.7
Female	20.8	41.2	32.3	37.4
Both	16.2	24.3	19.6	25.3
Total				
Male	11.4	14.1	18.0	18.7
Female	17.5	35.2	33.3	32.6
Both	13.7	21.1	19.8	22.7

Source: Census of India, 1981, Paper 2 [1983], as quoted in R M Sundaram [1987].

TABLE 24: LITERACY RATES, 1981

Segment	Total Population	Scheduled Castes	Scheduled Tribes
Total	36.23	21.38	16.35
Male	46.89	31.12	24.52
Female	24.82	10.93	8.04
Rural	29.65	18.48	14.92
Male	40.79	27.91	22.94
Female	17.96	8.45	6.81
Urban	57.40	36.60	37.93
Male	65.83	47.54	47.60
Female	47.82	24.34	27.32

Source: Census of India, 1981, Primary census abstract.

indicators reflecting the status of health care. There are significant regional variations in public efforts (as revealed by expenditures) and outcomes (as shown by infant mortality, life expectancy, etc). These are shown in Table 22.

In absolute terms, per capita expenditures are very low in the three states with the highest concentration of poverty—Uttar Pradesh, Bihar and Madhya Pradesh. These are also the states with the lowest life expectancy and highest infant mortality rates. However, there appears to be no systematic relationship between health expenditures and outcomes. Even the rank relationship between the input and output variables is not high. This is largely because life expectancy and infant mortality are a function of relative priority accorded to primary health care, and numerous other factors such as protected water, female education, immunisation, etc. Detailed break-up of health expenditures is not available state-wise.

Assuming that the relative share of various segments of health expenditure remained unchanged (which is by and large the case in most states), life expectancy and infant mortality in 1986 may be expected to be a function of initial expenditure levels in 1976, and the rate of growth of expenditure over the period 1976-86. The estimated equations are as follows:

$$\begin{aligned} \text{Log (LE 1986)} &= 3.3 + \\ &0.25^* \\ &(0.061) \text{ Log (PEH 1976-77)} \\ &- 0.066^* \text{ Log (I+r)} \\ &(0.78) \quad R^2 = 0.63 \end{aligned}$$

$$\begin{aligned} \text{Log (IMR 1986)} &= 6.2 \\ &- 0.64^* \\ &(0.3) \text{ Log (PEH 1976-77)} \\ &+ 2.4 \text{ Log (I+r)} \\ &(3.8) \quad R^2 = 0.39 \end{aligned}$$

* Statistically significant where

LE = Life expectancy.

PEH = Level of per capita expenditure on health (in different states).

IMR = Infant mortality rate.

The relationship between levels of expenditures on health and life expectancy is in the expected direction, although the rate of growth of expenditures is non-significant. The elasticity of life expectancy to expenditure was in the range 0.13-0.37. Infant mor-

ality rate, however, is weakly related to expenditure and not at all to its growth rate. Clearly, policies such as immunisation and nutrition interventions are more relevant in this context than aggregate expenditures.

(C) EDUCATION

In the field of education, there has undoubtedly been considerable progress in terms of the growth of institutions. Yet serious inequality in access to education persists by regions, social groups, and gender [Subbarao, 1987b]. Basic data showing these inequalities in access to education are given in Tables 23, 24 and 25. Even on the average, all-India literacy rate in 1981 stood at a low level of 36.2. Drop out rates are high and there is considerable wastage.

Despite a rise in government expenditure on education as a per cent of GDP, the quality, physical facilities, and geographical coverage of schools are poor especially in rural areas. Even though social returns for primary education are known to be higher, government spending is based in favour of higher education whose recipients are largely the rich and the elite. Cost recovery is poor at higher levels of education so that a substantial chunk of subsidy goes to the non-poor.

While the above remarks give the general status of education in India, there are significant regional variations as in other spheres of development. State-wise expenditures and literacy rates are shown in Table 22. Unlike in the case of health, literacy rates and educational expenditures are correlated:

$$\begin{aligned} \text{Literacy Rate} &= 1.352 + 0.82^* \text{ Eq Exp,} \\ R^2 &= 0.67 \end{aligned}$$

* Significant at the 5 per cent level.

Since educational outcomes are related to expenditures, it would be interesting to examine whether states with unsatisfactory performance are spending adequately on education. In order to do this we first computed the ratio of expenditure to per capita income for each state. We then ordered the states in terms of the distance of each state from the best in India. A relationship between the two series was estimated in order to know whether states are spending adequately, in relation to their perceived needs. This equation of literacy on effort yielded no statistically significant relationship, clearly confirming the underfunding on education in many states.

By far the most challenging problem that

India is now facing in education is to ensure that girls and children of poverty groups (e.g., scheduled castes and tribes) are not merely enrolled but retained in school after enrolment. Given the resource crunch, a major question is one of reconciling affordability with cost recovery—a problem common to all social sectors. Charging the poor full cost will certainly put the services beyond the reach of the poor. On the other hand, fully subsidising the services would place such a burden on public budget as to affect growth programmes. At present, cross-subsidising between the more affluent and the poor seems to offer a promising way out. For example, Rs 6,243 million was spent on higher and professional education during 1976-77. If 50 per cent of the cost were recovered, there would have been a saving of Rs 3,121 million, which, if diverted to primary education (assuming no constraints are on the demand side), could have provided primary education to about 20 million more children. If this were continued for five years (1976-81), the country would have had 100 million additional literates, or a literacy rate of 51.4 per cent instead of the realised 36.2 per cent in 1981.

VIII

Concluding Remarks

An important conclusion of this study is that the beneficial effects of growth on the incidence of poverty can be nullified by adverse movements in the inequality of consumption. Indeed, whether analysed at the state-level or all-India level, this has been the case during 1973-77 when high growth in average consumption per capita was accompanied by a sharp rise in consumption inequality, thereby considerably reducing the total impact on the reduction of poverty incidence. To this extent, the *policy response* in the form of a series of anti-poverty interventions since the mid-1970s aimed at raising the income/consumption levels of the poor and the ultra poor was basically a sound policy response.

During the period 1977-83, average consumption grew slowly, but consumption inequality fell in many states, and the reduction in the incidence of poverty was greater in magnitude than in the earlier period of high growth. Our separate analysis for the 'poor' and 'ultra poor' suggests that the beneficial impact of a reduction in inequality is more pronounced for the ultra

TABLE 25: RETENTION RATES (CLASS I-V)

Period	For all Communities	For Scheduled Castes	For Scheduled Tribes	For Other Communities
1968-72	33.5	27.8	20.3	35.3
1969-73	35.0	28.8	21.3	36.9
1970-74	36.2	30.3	22.9	38.0
1971-75	36.8	31.4	22.3	38.6
1972-76	37.2	32.2	21.9	39.1
1973-77	36.9	33.5	20.5	38.8
1974-78	38.6	35.6	23.3	40.5

Source: Ministry of Education, New Delhi.

poor than for the poor; by the same token a worsening of inequality hurts the ultra poor proportionately more than the poor. While it is difficult to identify precisely the factors that may have contributed to the decline in inequality in many states during 1977-83, the role of direct interventions cannot be minimised.

An interesting temporal tendency worth noting is that the poverty ratio has become more responsive to the changes in mean income and income inequality in all states except Bihar and West Bengal. Moreover, the numerical magnitudes of the relevant elasticities are higher for the richer states. Also, both growth and inequality elasticities are higher for the ultra poor than for the poor. These results show that it would seem desirable to pursue inequality-reducing policies in both rich and poor states to compound the beneficial impact of growth especially on the ultra poor.

These results lend significant credibility to the consumption-equalising interventions initiated since the mid-1970s. There is no doubt that at present there is significant inter-state variation in the effectiveness of these interventions which in general need to be improved. However, the divergence in the performance of programmes such as IRDP between the macro and micro level has been narrowing, and our review of micro evidence suggests substantial benefits even in a poor state such as Uttar Pradesh. While this is consistent with our earlier finding of substantial reduction of consumption inequality in this period, programme effectiveness is clearly weak in the poorer states. Employment programmes contributed substantially to *incremental* employment/income growth, and their state-wise distribution by and large corresponded with the distribution of the ultra poor, although poorer states (Bihar and West Bengal) deserved greater efforts.

Finally, our analysis of social sector (including nutrition) investments and performance suggests under-investment when efforts are analysed in relation to perceived needs and the capacity of the states. On the whole, the general record on health, education and nutrition fronts is most unimpressive. One is left with the impression that the social policies which can raise the long-run capabilities of the people have generally been relegated to the background in Indian policy-making.

Notes

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- 1 For a detailed discussion of this issue see Kakwani [1986].
- 2 Two consecutive years, 1972-73 and 1973-74, were selected with a purpose. As is well-known, 1972-73 was a bad crop year, and

1973-74 was a normal year. Food production in the first year was lower than in the second year by 8 per cent. We thought these two year data would provide a good opportunity for studying the short-run impact of crop output fluctuation on poverty and inequality across the states.

- 3 See for instance Kakwani [1980], Clark, Hemming and Ulph [1981], Foster, Greer and Thorbecke [1984] and Takayama [1979].
- 4 It is worth repeating that this procedure, unlike the procedures adopted by many previous researchers, appropriately takes account of the regional price variations while deriving poverty at the all-India level.
- 5 It is worth stressing that this finding contradicts the results based on cross-country comparisons by Fields.
- 6 Kakwani [1989] has provided the explicit expressions of this elasticity for all the additively decomposable poverty measures.
- 7 See Kakwani [1989].
- 8 To illustrate, consider a state which assisted households close to the poverty line and helped them all cross the line, with another state that selected households way below the poverty line (i.e., the ultra poor), and helped their incomes grow, but not sufficiently enough to cross the poverty line. Previous IRDP evaluations hailed the former state and condemned the latter, following the criterion of "crossing the poverty line". Gaiha's [1989] critique misses this important consideration in the assessment of programme effectiveness.

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